

# Protein Docking Score Prediction with SMILES Fingerprints

Zhangdaihong Liu  
Digital Health Group  
Oxford-Suzhou Centre for Advanced Research  
Suzhou, China  
Jessie.Liu@oxford-oscar.cn

Jingtong Jin  
Shenyuan Honors College  
Beihang University  
Beijing, China  
jingtong.jin@outlook.com

Chenyang Wang  
Department of Engineering Science  
University of Oxford  
Oxford, UK  
chenyang.wang@keble.ox.ac.uk

Yang Yang  
Department of Engineering Science  
University of Oxford, Oxford, UK  
Oxford-Suzhou Centre for Advanced Research, Suzhou, China  
yang.yang@eng.ox.ac.uk

David A. Clifton  
Department of Engineering Science  
University of Oxford  
Oxford, UK  
david.clifton@eng.ox.ac.uk

**Abstract**—In this paper, we explain how we solve the IEEE COVID-19 Bioinformatics Drug Target Challenge. The goal is to predict the docking scores of 18 proteins for over 30,000 unseen drug molecules using their SMILES sequences. We embed SMILES with structural molecule fingerprints and compare four types of fingerprints produced by commonly used toolkits. Model-wise, we focus on two non-linear models, XGBoost, a gradient tree-based ensemble learning method, and a feed-forward neural network (FFNN). We use XGBoost to determine the types of fingerprints and set benchmark results, and use FFNN to report the final results on the test set since it has superior performance.

## I. INTRODUCTION

### A. Task description

The task is to build machine learning models to predict docking scores of 18 SARS-CoV-2 protein targets with drug candidates provided in the canonical SMILES form. The training dataset has 270,000 SMILES sequences and their docking score for 18 target proteins. The external test set which requires predictions for has 30,457 SMILES sequences. The only evaluation criterion is the mean absolute error (MAE) between the predicted and true docking scores on the given test set.

### B. Background

Virtual screening, a computational strategy for molecule screening, has been extensively employed in the pharmaceutical industry for high-throughput drug identification recently [2, 10]. Molecular fingerprints are commonly used for this *in silico* procedure to transform chemical structures into scalar vectors. Reproducing as much the molecule’s properties as possible while keeping relatively low dimensions for higher efficiency, fingerprints can be good representations for computing the docking score, a qualitative indicator showing how a molecule fits with a receptor protein [5, 10]. Therefore, good fingerprint representations of molecules can help to predict original molecules’ drug-likeness.

### C. Our work

In this work, we transformed SMILES into multiple fingerprints as our models’ inputs. We compared several basic machine learning models including linear regression, AdaBoost, random forest, SVM and XGBoost. We also built feed-forward neural network models whose architectures were decided by random search. We compared the MAE of all considered models on our internal test set which was randomly separated from the given training data of 270,000 molecules. All experiments were performed using 5-fold cross-validation. The best-performed model was then applied to the test set for predicting the final docking score.

## II. METHODS

### A. Embed SMILES: Fingerprints

We compared 4 different commonly used structural molecular fingerprints, including Molecule Access System (MACCS) fingerprint [3], RDKit fingerprint [8], Extended-connectivity fingerprint (ECFP) [9] and Min-hash fingerprint (MHFP) [7]. A detailed description is provided below and concluded in Table I.

1) *MACCS fingerprint*: MACCS fingerprints, one of the most commonly used fingerprints, has bits representing the presence in the molecule of certain substructures (e.g., the presence and absence of the fingerprint is denoted as 1 and 0, respectively). There are different sizes of MDL key sets (a commonly used molecule database for searchable substructures); however, only the one with 166 keys is publicly available. We used RDKit toolkit [8] to obtain 167 bits, where one extra unused bit is added to preserve the MACCS’s key indices.

2) *RDKit fingerprint*: An RDKit-specific fingerprint is a path-based topological fingerprint identifying molecule subgraphs given a pre-defined range of sizes, including individual bond information based on bond type, atom type and atom degree. We adopted the default setting where the number of

bonds to include in the sub-graphs ranges from 1 to 7 and the output length is 2048 bits.

3) *ECFP*: ECFP is a type of circular fingerprint, encoding substructures within certain ranges of each atom’s neighbourhoods. We adopted a setting where the neighbourhood’s maximum diameter is set to 4 (also known as ECFP4) with an output length of 2048 bits.

4) *MHFP*: MHFP is a state-of-the-art molecule fingerprint re-implementing ECFP with a locality sensitive hashing. The MHFP we adopted in this work has the initial neighbourhood diameter set to 6 with an output length of 2048 bits.

TABLE I  
FINGERPRINTS CHARACTERISTICS

| Fingerprint | Method              | Length | Type   |
|-------------|---------------------|--------|--------|
| MACCS       | Fixed substructures | 167    | binary |
| RDKit       | Path-based          | 2048   | binary |
| ECFP        | Circular            | 2048   | binary |
| MHFP        | Circular            | 2048   | dense  |

### B. Final Method

After comparing a few different models and fingerprints (see Section II-D for more details), we selected ECFP as our final predictive feature and feed-forward neural network (FFNN) as the class of models to report the test results. Fig. 1 illustrates our method pipeline. We firstly converted SMILES sequences to ECFP (fingerprints) and then fed these fingerprints to FFNN. We adopted multi-layer perceptron (MLP) and used random search to determine the model architecture. In the random search, we mainly considered the following hyper-parameters: layer depth, layer width and dropout rate. For each of the proteins (columns), we fitted one MLP and did 20 random searches. In particular, the ECFP fingerprint has a dimensionality of 2048 and the final output is the predicted docking score for one protein. In the random search, we set the maximum number of hidden layers to be 8. The layer depth is randomly chosen first and the layer widths are then generated to equally divide the interval [2048, 1]. We considered 3 dropout rates, 0, 10% and 20%.

The predicted docking scores on the external test set were obtained by firstly converting the SMILES sequences to the ECFP fingerprints, and then feeding the fingerprints into the optimal neural network model trained for the corresponding protein.

### C. Implementation details

The provided training data was split into 5 equal-sized sets randomly. We left one of the 5 sets out as our *internal test set* and use the rest 4 sets (*internal training set*) for random search and tuning models. Notably, this split is fixed for all our experiments. In the random search, each converged model from the internal training set was applied to the internal test set to calculate a MAE value. Model with the smallest MAE was then applied to the corresponding column in the *external test set* to give the final results. We implemented the neural

network model with Pytorch version 1.5.1 (locally) and 1.4.0 (on cluster) [6], and used AdamW [4] as the optimiser, mean-squared error as loss function and ‘ReduceLROnPlateau’ with patience being 2 as the learning rate scheduler. We also applied an early stopping criterion: the 4 sets of data were further split into 5 folds with 4 folds as training data and 1 fold as validation data. The algorithm stops if the validation loss is not smaller than the current minimum loss for 5 consecutive epochs.

The code can be found in the Github repository: <https://github.com/lzdh/IHS-H2-Bioinformatics-Drug-Target>.

### D. Other baseline methods

We also tried several other methods including linear regression, random forest regression, support vector regression and XGBoost regression (XGB; a regularizing gradient boosting framework [1]). Among these methods, XGB gave the best results on the first try and therefore, we focused on fine-tuning XGB. Similarly to the neural network model, we fine-tuned one XGB model for each protein and focused on tuning 3 main parameters: the number of estimators, the depth of the tree and the minimum child weight. Fine tuning XGB is very time-consuming on large dataset. Therefore, we fine-tuned XGB on a few randomly selected proteins and took the setting that was returned as the optimal setting for the most iterations to apply to all proteins. This setting has the number of estimators set to 50, depth of the tree 20 and minimum child weight 15. Other model parameters can be found in our Github repository.

We used the results from XGB to be our benchmarks and select fingerprints to be used in the neural network.

## III. RESULTS

### A. Analysis of fingerprints characteristics

Fingerprints for the train set were projected using UMAP in the 2D space shown in Fig.2. The visualisations show a great difference between MACCS and the other three types, as MACCS’s projection displaying a few dense clusters. This result is expected since the dimensionality of MACCS (167) is significantly smaller than the other three (2048). This reduction in dimensionality may decrease the ability of the fingerprints to capture data structures and resulting in overlapping compact clusters. For the other three types with the dimensionality of 2048, the RDKit fingerprint is the most evenly distributed, and ECFP is similar to MHFP but more scattered. The dispersion of RDKit fingerprint in the UMAP projection may attribute to its path-based algorithm. The similarity between ECFP and MHFP can be explained by the same implementation of finding circular substructures. Further comparison between the four types of fingerprints will be discussed in the following sections.

### B. XGB point estimation on the four fingerprints

Table II compares the MAE obtained by XGB using the four fingerprints. The averaged MAEs for MACCS, RDKit, ECFP and MHFP are 0.3905, 0.3867, 0.3652 and 0.3958, respectively. It is observed that ECFP has the lowest MAE

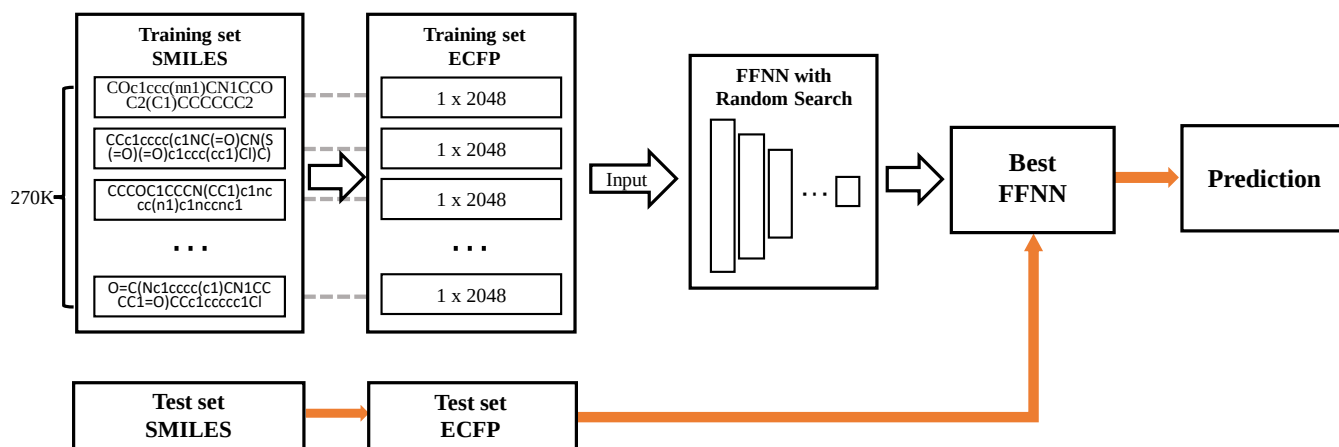


Fig. 1. Pipeline of the final method used to produce the predictions.

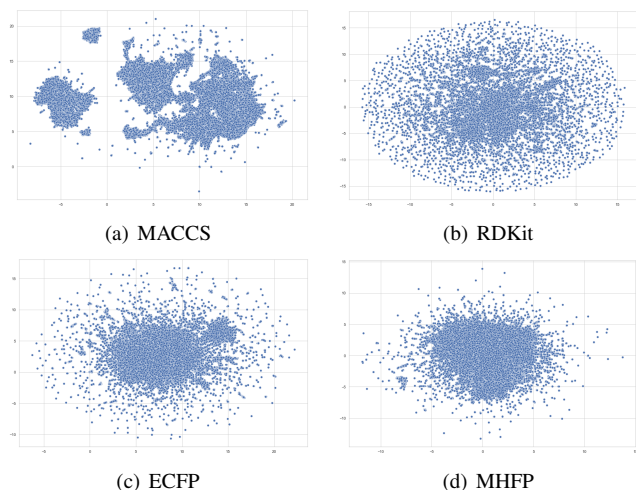


Fig. 2. Fingerprints UMAP Visualization. The UMAP projection of 4 types of fingerprints on the training set. Every dot represents a single drug candidate molecule. Axes are dimensionless and come from UMAP dimension reduction.

for all 18 proteins. Referring back to Fig. 2, the visualisation for ECFP has more distinctive clusters compared with the other three. Therefore, we chose ECFP for the further neural network experiments.

### C. Neural network results

We obtained 4 unique neural network architectures through the random search on the 18 proteins (Table III), with the dropout rate all being 0. The MAE on the internal test set is shown in Table IV, and the averaged MAE for the 18 proteins is 0.3330. Notably, these results were derived from using ECFP as input. Compared with the XGB, neural network models offered lower MAE on every single protein (Table IV).

Normalised by the range of the true internal test labels (the range from minimum to maximum), we found protein 3CL-Pro\_pocket1 has the largest MAE value and COV\_pocket10

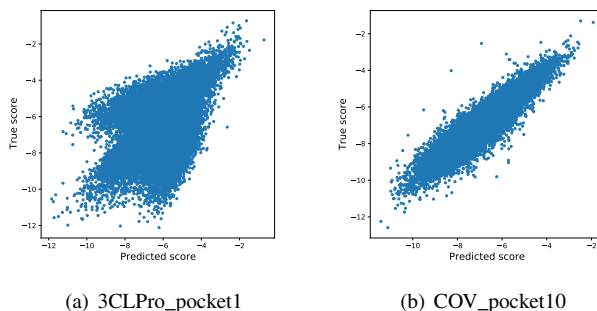


Fig. 3. The predicted scores versus the true scores for the protein with the largest (a) and smallest (b) normalised MAE.

has the smallest MAE (‘MAE\_norm’ rows in Table IV). Their predicted and true scores are shown in Fig. 3.

## IV. CONCLUSIONS

In this challenge, we attempted to predict protein docking scores using the SMILES sequence of drug molecules. Producing informative SMILES sequence representations is a key to improve accuracy of prediction scores. Through literature review, we employed four sets of molecule fingerprints using four widely applied molecule structure converting methods. Although the MHFP was reported to have superior performance in literature [7], via comparison in our work, ECFP offered the best results among all the fingerprints we considered. ECFP also has better-structured visualisation.

Furthermore, we built one machine learning model for each target protein to minimise the MAE. Several baseline models were compared to give the benchmark results. Among the baseline models we examined, XGB gave the best result. We further fine-tuned XGB by adjusting the maximum tree depth, the minimum child weight and the number of estimators, which improved the MAE. We then attempted the neural network models, in particular MLP, which gave better results than XGB. To further reduce the MAE, we applied random

TABLE II  
XGB POINT ESTIMATION RESULT (MAE) ON THE INTERNAL TEST SET BY THE FOUR FINGERPRINTS AND FOR ALL 18 PROTEINS.

| Fingerprint | 3CLPro_pocket1 | ADRP-ADPR_pocket1 | ADRP-ADPR_pocket5    | ADRP_pocket1          | ADRP_pocket12 | ADRP_pocket13  |
|-------------|----------------|-------------------|----------------------|-----------------------|---------------|----------------|
| MACCS       | 0.8042         | 0.4482            | 0.4492               | 0.3051                | 0.3051        | 0.3731         |
| RDKit       | 0.8037         | 0.4374            | 0.4371               | 0.3085                | 0.3058        | 0.3625         |
| ECFP        | <b>0.7544</b>  | <b>0.4298</b>     | <b>0.4306</b>        | <b>0.2882</b>         | <b>0.2877</b> | <b>0.3501</b>  |
| MHFP        | 0.8223         | 0.4605            | 0.4626               | 0.3118                | 0.3120        | 0.3810         |
| Fingerprint | COV_pocket1    | COV_pocket2       | COV_pocket8          | COV_pocket10          | NSP9_pocket2  | NSP9_pocket7   |
| MACCS       | 0.2325         | 0.2308            | 0.3773               | 0.3909                | 0.4237        | 0.3461         |
| RDKit       | 0.2576         | 0.2553            | 0.3751               | 0.3768                | 0.4112        | 0.3384         |
| ECFP        | <b>0.2121</b>  | <b>0.2109</b>     | <b>0.3502</b>        | <b>0.3649</b>         | <b>0.3865</b> | <b>0.3333</b>  |
| MHFP        | 0.2435         | 0.2416            | 0.3824               | 0.3927                | 0.4194        | 0.3578         |
| Fingerprint | NSP15_pocket1  | ORF7A_pocket2     | PLPro_chainA_pocket3 | PLPro_chainA_pocket23 | PLPro_pocket6 | PLPro_pocket50 |
| MACCS       | 0.3666         | 0.3285            | 0.3569               | 0.4398                | 0.3647        | 0.4859         |
| RDKit       | 0.3578         | 0.3287            | 0.3582               | 0.4346                | 0.3382        | 0.4730         |
| ECFP        | <b>0.3443</b>  | <b>0.3075</b>     | <b>0.3292</b>        | <b>0.4108</b>         | <b>0.3313</b> | <b>0.4518</b>  |
| MHFP        | 0.3692         | 0.3300            | 0.3563               | 0.4419                | 0.3559        | 0.4835         |

TABLE III  
OPTIMAL NEURAL NETWORK ARCHITECTURE FOR THE 18 PROTEINS OBTAINED VIA RANDOM SEARCH.

| Neural network architecture                | Protein                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------|
| [2048, 1752, 1460, 1168, 876, 584, 292, 1] | ADRP_pocket12, PLPro_pocket6, NSP9_pocket2, PLPro_pocket50                                          |
| [2048, 1705, 1364, 1023, 682, 341, 1]      | 3CLPro_pocket1, ADRP-ADPR_pocket5, ADRP_pocket13, COV_pocket8, NSP9_pocket7                         |
| [2048, 1636, 1227, 818, 409, 1]            | ADRP_pocket1, COV_pocket1, COV_pocket10, ORF7A_pocket2, PLPro_chainA_pocket3, PLPro_chainA_pocket23 |
| [2048, 1536, 1024, 512, 1]                 | ADRP-ADPR_pocket1, NSP15_pocket1, COV_pocket2                                                       |

TABLE IV  
NEURAL NETWORK RESULTS COMPARED WITH THE BEST XGB RESULTS USING ECFP FOR THE 18 PROTEINS ON THE INTERNAL TEST SET. MAE\_NORM IS CALCULATED BY THE MAE VALUE NORMALISED BY THE RANGE OF THE TRUE SCORES. THE LARGEST AND SMALLEST VALUES ARE NOTED BY BOLD FONT.

| Protein  | 3CLPro_pocket1 | ADRP-ADPR_pocket1 | ADRP-ADPR_pocket5    | ADRP_pocket1          | ADRP_pocket12 | ADRP_pocket13  |
|----------|----------------|-------------------|----------------------|-----------------------|---------------|----------------|
| XGB      | 0.7544         | 0.4298            | 0.4306               | 0.2882                | 0.2877        | 0.3501         |
| FFNN     | 0.7456         | 0.3468            | 0.3506               | 0.2632                | 0.2612        | 0.2941         |
| MAE_norm | <b>0.0655</b>  | 0.0297            | 0.0291               | 0.0351                | 0.0348        | 0.0290         |
| Protein  | COV_pocket1    | COV_pocket2       | COV_pocket8          | COV_pocket10          | NSP9_pocket2  | NSP9_pocket7   |
| XGB      | 0.2121         | 0.2109            | 0.3502               | 0.3649                | 0.3865        | 0.3333         |
| FFNN     | 0.2055         | 0.2062            | 0.3156               | 0.3127                | 0.3542        | 0.2808         |
| MAE_norm | 0.0433         | 0.0429            | 0.0336               | <b>0.0277</b>         | 0.0334        | 0.0289         |
| Protein  | NSP15_pocket1  | ORF7A_pocket2     | PLPro_chainA_pocket3 | PLPro_chainA_pocket23 | PLPro_pocket6 | PLPro_pocket50 |
| XGB      | 0.3443         | 0.3075            | 0.3292               | 0.4108                | 0.3313        | 0.4518         |
| FFNN     | 0.3130         | 0.3017            | 0.3250               | 0.3938                | 0.2989        | 0.4131         |
| MAE_norm | 0.0316         | 0.0390            | 0.0427               | 0.0379                | 0.0305        | 0.0336         |

search and found 4 different optimal architectures for the 18 proteins. In the end, the optimal neural network model was used to give the prediction scores on the test set for each protein.

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